Markets in Series

Filtering, belief propagation, and arbitrage as the loop correction

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Abstract

Parallel composition of cost-function market makers is settled algebra: merging is infimal convolution, liquidities add, and for quadratic makers the merge is Gaussian precision fusion exactly. This note develops the serial operator. Factorize a joint distribution by the chain rule and run one market per conditional factor, each opening when its conditioning information freezes and settling in cascade through the graph. For linear-Gaussian structure the correspondence with inference is exact: alternating a model step with a market step reproduces the Kalman filter, because the market step is the Kalman update, and market messages on a tree compute exact posterior marginals, belief propagation with trades as messages. On loopy graphs the pathology of belief propagation materializes as arbitrage: pairwise quotes around a cycle that admit no joint distribution form a matrix with a negative eigenvalue, handing any trader a bundle with negative price and nonnegative payoff, so arbitrageurs are the loop correction. The two operators are the two operations of the inference semiring, parallel the product and serial the sum, and exactness for Gaussians has a single source: the family on which sum-product and min-sum agree. The prior art is heavy and credited; the serial architecture, the market-implemented filter, and the loopy-arbitrage certificate appear to be the unoccupied residue.

1 Two operators, one semiring

Merging market makers that quote the same quantity is infimal convolution of their costs: conjugates add, liquidity adds (Bhaskara et al. 2023), and the composition law is the risk-sharing algebra (Barrieu and El Karoui 2005). For quadratic makers the merged clearing price is the precision-weighted mean of quotes and the merged price impact the inverse of summed precisions: parallel composition is Gaussian fusion. This note is about the other operator. A model is a factorization,

$$P(\text{everything}) \;=\; \prod_{\text{nodes}} P(x_i \mid \text{parents}(x_i)),$$

and serial composition runs one market per factor, each pricing its conditional given what is upstream.

That there are exactly two operators is not an accident of taste. Message-passing inference is an algorithm over a commutative semiring (Aji and McEliece 2000): one operation combines evidence about a variable (the product), one moves evidence between variables (the sum), and sum-product, max-product and min-sum are the one algorithm over different semirings. The market algebra is that pair: parallel merge is the product, densities multiplying as precisions add, and serial propagation is the sum. The parallel operation wears two costumes, addition of conjugates and infimal convolution of costs, because inf-convolution is convolution in the min-plus semiring and the Legendre transform is its Laplace transform (Litvinov 2005): the convex machinery of this and the companion notes is the tropical shadow of the probability calculus. Exactness in everything Gaussian below has one source: log-quadratics are the family on which the sum-product and min-sum semirings coincide, so a market whose traders optimize agrees with an inference engine that integrates. Off the family they diverge by the Laplace-approximation gap, which is §6's open question.

2 One market per factor

The locality that makes the factorization tradable is Hanson's modularity: in a combinatorial LMSR a bet on $A \mid B$ moves that conditional and provably nothing else, uniquely among market scoring rules (Hanson 2007, 2003). Hanson runs one joint market over the product space; the substance of pricing it is probabilistic inference, which is why exact pricing is #P-hard (Chen, Fortnow, et al. 2008), why tournament markets price by Bayes-net inference (Chen, Goel, et al. 2008), why a deployed combinatorial market ran its price and asset updates on the junction-tree algorithm (Sun et al. 2012), and why approximate designs price over the marginal polytope (Dudík et al. 2012; Dudík et al. 2021). Securities structured by a Bayes-net factorization appear in Pennock and Wellman (2000), with the trades that preserve the structure characterized by Xia and Pennock (2011).

The serial architecture takes the factors as separate venues rather than one joint book. Market i opens on $P(x_i \mid \text{parents})$ when its conditioning information freezes, conditions on upstream quotes while upstream is live, and re-references when upstream settles; settlements cascade through the DAG like dataflow, and the schedule of markets is an unrolling of the graph. Observable nodes carry settled markets; a latent node either settles through its observable footprint, a market on a hidden state being a market on a functional of future observables scored through the model, or remains indicated but unsettled, priced by whoever warehouses the model risk.

3 The market step is the Kalman update

Take the linear-Gaussian chain $x_{t+1} = ax_t + \varepsilon_t$, $y_t = x_t + \eta_t$, and alternate two steps: a *model step*, propagating the current belief through the dynamics in mean-precision form, and a *market step*, merging the propagated belief with an observation maker quoting y_t whose capital is the observation precision.

Proposition 1. *The alternation is the Kalman filter (Kalman 1960): after each observation the market state equals the filtered mean and variance exactly.*

Proof. The model step is the standard prediction of mean and variance. By the fusion identity, merging quadratic makers adds precisions and precision-weights means, which is the information-form measurement update; the covariance-form update follows by the usual algebra. ■

Prediction is computation; correction is a market; capital decides how hard the data pulls. The nearest published object is the Bayesian market maker of Brahma et al. (2012), whose trade update is a scalar Gaussian measurement update with covariance inflation for jumps; the filtering reading, and any alternation with a dynamic model, appear unstated in the literature.

4 Trees: messages are trades

On a Gaussian chain with an observation at every node, the posterior at an interior node is the merge of three sources: the forward message (the filter of §3 run up to the node), the local observation, and the backward message (downstream observations pulled back through the dynamics, each pull-back a reparametrization and each combination a precision addition).

Proposition 2. *All three messages are market operations, and the merged posterior equals the brute-force conditioning of the joint at every node: Gaussian belief propagation (Pearl 1988) with trades as messages, exact on trees.*

The companion repository verifies the identity to ten digits. The contrast with the deployed combinatorial engines is architectural: there the junction tree is the pricing algorithm inside one joint market (Sun et al. 2012); here the messages pass between venues, and the graph's edges are market boundaries.

5 Loops: cycle inconsistency is arbitrage

On loopy graphs belief propagation double-counts and its fixed points drift. The market version of the pathology is sharper and self-punishing.

Proposition 3. *Assemble pairwise correlation quotes around a cycle into a quote matrix P with unit diagonal. The quotes admit a joint distribution iff $P \succeq 0$. Otherwise, with w the eigenvector of a negative eigenvalue λ, the bundle with weights $ww^\top$ has price $w^\top P w = \lambda < 0$ and payoff $(w^\top x)^2 \geq 0$: a sure profit of at least $|\lambda|$ per unit.*

This is the second-moment coherence condition, prices of products must form a PSD matrix (d'Aspremont 2005), read as cycle consistency. The lineage of "arbitrageurs enforce coherence" is long and must be owned: coherence is no-arbitrage (Nau and McCardle 1991); Pennock and Wellman (1996) built arbitrageur agents enforcing the additivity identities of a Bayes-net economy, with equilibrium prices equal to the network's probabilities and distributed bidding as distributed inference; the combinatorial literature removes the arbitrage algorithmically over the marginal polytope (Kroer et al. 2016); and Polymarket's arbitrageurs have been measured doing the enforcement for eight figures (Saguillo et al. 2025). What the loop reading adds is the object being corrected: the inconsistency that degrades loopy belief propagation is, in a market, not a numerical nuisance but free money, and the flow that harvests it pushes the quotes back toward the cone. Whether the corrected fixed point is the true marginal, a Bethe-like surrogate, or something the liquidity profile selects is open.

6 Discussion

Three structures elsewhere in this project are serial compositions: boosting is sequential residual markets, each stage trading what the composite so far got wrong; a microprediction supply chain is a sequence of markets each adding value to an intermediate product; and Fermi-style decomposition of a question into a chain of small markets is the factorization run by hand. The equilibria-as-graphical-models correspondence is older than all of them (Pennock and Wellman 1996; Storkey 2011); what the serial operator contributes is the mechanism version: separate venues, an opening rule, cascading settlement, and exactness theorems where the Gaussian family permits them.

The open question underneath is the semiring gap. A market of optimizing traders natively computes min-plus, so off the Gaussian family it prices the max-product posterior rather than the sum-product marginal, and the discrepancy is the Laplace-approximation gap. Risk aversion is an exponential tilt, which suggests it interpolates between the two semirings; whether a risk-averse trading population prices the marginal, the mode, or a temperature in between is the right next theorem, and it would say precisely which inference algorithm a market is.

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